

Museums, Computer History and Institutional Challenges

SIGGRAPH 2021 Retrospective Panel

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1 INTRODUCTION

This panel is dedicated to exploring the challenges that face the exceptional cultural institutions and museums that collect the hardware and software of computer history and computer art history. The panel has three presentations by active practitioners working at some of the most prestigious institutions in the field worldwide:

1.1 “The Material Culture of Human-Computer Interaction: Highlights at the National Museum of American History” by Alana Staiti

This presentation will provide an overview of objects in the collections of the Smithsonian National Museum of American History related to real-time graphics and human-computer interaction. From one-of-a-kind systems of the early years of digital computing to prototypes of the first home entertainment systems to innovative adaptive technologies for people with disabilities, the Museum’s collections reveal a range of objects that tell a rich and layered history of technological innovations in human-computer interaction as well as the collecting agendas of museum curators past and present. We will conclude with a brief discussion of the future of collecting the history of computer graphics and interactivity at the Museum, including the challenges that lie ahead.

1.2 “Our Digital Fragility. The Preservation of Computer-Based Artworks” by Margit Rosen

The collection of the ZKM comprises one of the largest holdings of computer-based artworks. These complex digital objects reveal a dilemma of two centuries whose key technologies are electronics and the digital: The material culture of the 20th and 21st centuries is exceedingly fragile. Hardware disintegrates within a short period of time, whether it is used or not. Companies that manufactured certain devices or elements disappear. And if the artwork is connected to the Internet, it becomes a victim of rapidly changing protocols and standards. The general problem of preserving our digital culture becomes particularly acute from the perspective of a museum. The contradiction between a technical industry designed for change and works of art intended for eternity cannot be resolved. The ZKM is one of the few institutions worldwide that has been collecting computer-based works since the 1990s and has therefore had to develop preservation strategies. The lecture will address the question of how we can meet the challenges of preservation of complex digital objects and whether in 100 years we will have a chance to walk through museums and experience alongside paintings, graphics, and sculptures also computer-based artworks – works of art produced in the media that defined the everyday life of their time.

1.3 “Preserving the Bumpy History of Multimedia and its Roots in Hypermedia Dreams” by Marc Weber

This presentation uses the up and down history of multimedia and hypermedia as a lens into the collection and exhibits at the Computer History Museum, as well as to examine broader issues of preservation and access. If they think of it at all, people mostly imagine the history of multimedia as a smooth progression upward, from crude line-printed ASCII art to today’s eye-wateringly crisp 3D games and streaming video. All in time to a thumping baseline from wireless subwoofers. Few think of hypertext and hypermedia – the core navigation model for the online world – as integral to that history. That’s partly because the Web tore down most of the multimedia advances of the 1980s, its slow connection speeds and poor protocols pushing back the quality of sound, graphics, and video a decade or more. Only with broadband and more efficient file formats (Flash, MP3, etc.) did Web media begin to claw itself back the level of standalone PCs in the late 1980s. Yet this was far from the first time that advances in communication had been at odds with ways of representing media. From its invention through the

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printing press, the telegraph, and the early computer, writing was easier to copy than images. From its start in the early 20th century, hypertext pioneers were as concerned with navigating different media as with text. This led to some of the klugey solutions we cover at the Computer History Museum, like computer controlled microfilm readers (PLATO) and videodisc players (Aspen Movie Map).

2 BIOGRAPHIES

Alana Staiti. Alana Staiti, Ph.D., is curator of the computing and information sciences collection at the Smithsonian National Museum of American History. Her research interests include the history of computer graphics and animation and 3D human modeling.

Margit Rosen. Margit Rosen is Head of the Departement Collection, Archives and Research at the ZKM | Center for Art and Media Karlsruhe. Her work as a curator and researcher is centered on the art of the 20th and 21st centuries, especially on the question of art's relationship to technological and political change. One focus of her research is the history of the appropriation of computer technologies by the arts since the 1950s. She has lectured extensively at various institutions, including Princeton University, Centre Pompidou, the Louvre, Deutsches Museum, and the Venice Biennale. Among her numerous publications may be mentioned: "A Little-Known Story about a Movement, a Magazine, and the Computer's Arrival in Art," MIT Press/Cambridge/Mass., 2011.

Marc Weber. Marc Weber is curatorial director of the Internet History Program at the Computer History Museum (CHM). He pioneered Web history as a topic starting in 1995, with crucial help from the Web's main inventor Sir Tim Berners Lee and other pioneers. He co-founded two of the first organizations in the field, and has curated a dozen exhibits or galleries at CHM on networking, mobile, and AI. He speaks and publishes widely, and presents and consults on related topics to major media, patent firms, and filmmakers. He was a software and biomedical consultant for over a decade in Silicon Valley and near Geneva, Switzerland. Weber has taught at the University of California and holds bachelors degrees in Neurobiology and in Creative Writing from Brown University.

Moderator - Andres Burbano. Andres Burbano is the SIGGRAPH 2021 Retrospective chair; he is a media artist and scholar and is an Associate Professor in the Department of Design at Universidad de los Andes, in Bogota, Colombia. Burbano holds a PhD in Media Arts and Technology from the University of California Santa Barbara; and is visiting professor at the Danube University in Krems, Austria, and the University of Applied Sciences in Potsdam, Germany. Burbano, originally from Colombia, explores the interactions of science, art, and technology in various capacities: as a researcher, as an individual artist, and in collaborations with other artists and designers. Burbano's work ranges from documentary video (in both science and art), sound and telecommunication art to the exploration of algorithmic cinematic narratives. The broad spectrum of his work illustrates the importance, indeed the prevalence, of interdisciplinary collaborative work in the field of digital art.